

DIRECTED ENERGY WEAPONS: High-Power Microwaves



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Parts 1 and 2 covered particle beam weapons and their technology, respectively. In this part we look at high-power microwaves (HPMs).

In the era dominated by electronic warfare and smartweapons systems facilitated by advances in electronics technologies, military planners and strategists have always been on the look out to acquire improved capabilities in countering artillery fire, providing self-protection to aircraft and credible ship defence against cruise missiles, destruction or disruption of command and control assets, suppression of enemy air defence systems, space control and security and so on.

HPM weapon systems with their speed-of-light delivery, all-weather capability to destroy an adversary's electronics systems, area coverage of multiple targets, minimum collateral damage, simplified tracking and beam pointing and a deep magazine address most of the requirements of military commanders of the day.

HPM weapons generate an intense blast of microwave energy strong enough to overload electrical circuitry, inducing currents large enough to temporarily disrupt electronic systems or permanently damage integrated circuits, causing these to fail minutes, days or even weeks later. The microwave blast may even melt the circuitry in some cases. Humans exposed to the blast of an HPM weapon remain unharmed and might not even know that they have been hit.

A number of technological challenges need to be addressed to come to the level of a deployable HPM weapon possessing

all of the above features. These include development of compact high-peak-power or average-power HPM sources, compact high-gain ultra-wideband antennae, compact and efficient pulse power drivers, predictive models for HPM effects and lethality to create a comprehensive database on effects of such a radiation on a wide range of land, sea, air and space based military assets and system integration to meet the requirements of a variety of military platforms such as fixed-wing and rotary aircraft, land vehicle, aircraft pods, naval combatants, unmanned aerial vehicles and so on.

An HPM weapon essentially comprises a pulse power source that drives the microwave source, HPM source and a transmitting antenna that directs the microwaves towards the target and acts as an interface between the microwave source and the atmosphere.

Besides these, HPM weapons also include tracking, aiming and control systems. The microwave power generated by the weapon, beam characteristics and target vulnerabilities together determine the effective operational range of the HPM weapon system.

The weapon is designed to disrupt, degrade or destroy electronics of the target by radiating electromagnetic energy in the microwave frequency band. Microwave frequencies range from 300MHz to 300GHz. HPM weapons under development occupy the frequency range of 500MHz to 3GHz. Frequencies around 100GHz have been exploited to build non-lethal weapons due to the significant effects this frequency has on human beings. This is primarily due to the coupling effects and generic electronic system vulnerabilities.

HPM weapons are broadly categorised as narrow-band HPM and ultra-wide-band HPM. While narrow-band HPM weapons radiate all their energy within few per cent

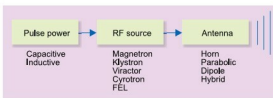


Fig. 1: Components of an HPM weapon system